
Averaged Jacobian Regularity Does Not Order Few-Step Error of Gaussian Interpolant ODEs

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Few-step sampling of a continuous-time generative model requires a probability
2 path, a solver, and a budget of field evaluations (NFE). Chen et al. propose mini-
3 mizing averaged squared Jacobian regularity A_2 as a path-selection criterion. We
4 ask whether the order that this scalar induces between two interpolants is reliable
5 at fixed NFE. Already for independent $N(0, 1)$ and $N(0, 4)$, the exact integrals are
6 $5\pi/8 - 1$ (linear) and $\pi^2/16$ (trigonometric VP), while Heun at NFE 8 reverses
7 Gaussian W_2 . The linear factor is the rational $6797469/3559400$; a rational en-
8 closure in $\mathbb{Q}[\pi, \sqrt{2}]$ gives $r_{\text{VP}} < 187/100$. Three regimes then separate. As a
9 pairwise comparator, linear versus VP invert in 5 of 12 geometry $\times$ solver cells (4
10 of 12 at every NFE in $\{8, 16, 32\}$). As an in-family objective, trigonometric VP
11 versus the scalar log-covariance schedule (Chen Ex. 3.3 with $M = \lambda_{\max}$) invert in
12 9 of 36 blocks and 4 of 12 cells. The unconstrained per-mode minimizer attains
13 the smallest $\mathcal{R}$ and the smallest W_2 in 36 of 36 blocks, but is not a shared (α, σ)
14 interpolant for $d \geq 2$.

15 1 Introduction

16 Few-step sampling commits to a path, a solver, and a small NFE budget before any error is observed.
17 Lipschitz constants enter classical one-step bounds [Hairer et al., 1993]. Chen et al. [2025] propose
18 minimizing averaged squared drift Lipschitzness (their Definition 3.2, denoted A_2) as a *selection*
19 criterion. They prove no bound relating A_2 to a discretization error. We test whether the pairwise
20 order that A_2 induces between two admissible paths is reliable at equal NFE.

21 We isolate that ranking question on commuting Gaussian interpolants, where the field is affine,
22 Gaussian W_2 is closed-form, and the regularity integrand is known exactly. The leading result is a
23 deterministic scalar Heun example (Section 3). A modal identity then records the signed information
24 that $\mathcal{R}$ discards (Section 4). Section 5 splits the ranking question into three regimes: pairwise linear
25 versus VP, in-family VP versus the shared Chen Ex. 3.3 scalar schedule, and the unconstrained
26 per-mode minimizer, which is not a shared (α, σ) interpolant for $d \geq 2$.

27 Contemporaneous work studies Lipschitz Euler rates [Stéphanovitch, 2026], sharpness-aware
28 grids [Gupta et al., 2026], least-action paths [Du et al., 2026], and instance-aware discretizations [Yuan
29 et al., 2026]. Those results control other objects. Tao and Choi [2026] give the general strain/vorticity
30 form of the phenomenon; Section 6 records the precise relation.

31 2 Gaussian flows and averaged regularity

32 Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent, and let (α, σ) satisfy $\alpha(0) = \sigma(1) = 1$
33 and $\alpha(1) = \sigma(0) = 0$. The interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023]

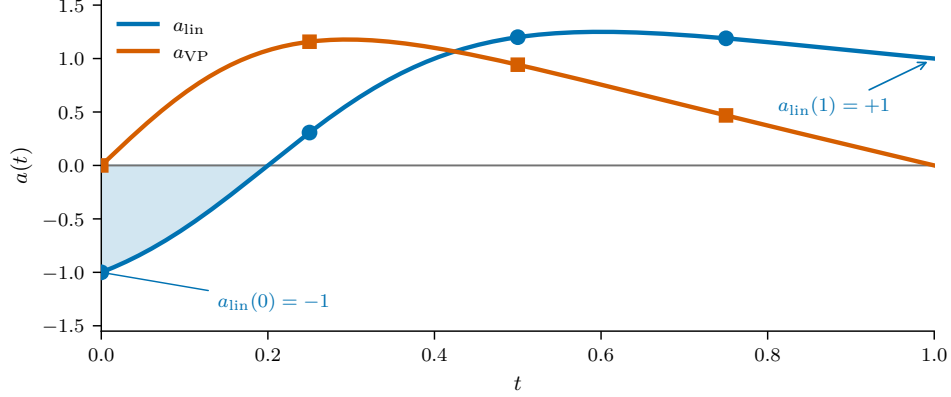


Figure 1: Scalar drifts $a_{\text{lin}}(t)$ and $a_{\text{VP}}(t)$ for independent $N(0, 1)$ and $N(0, 4)$. The shaded region is $a_{\text{lin}} < 0$. Markers are the four Heun nodes at NFE 8. $a_{\text{lin}}(0) = -1 < 0 < a_{\text{lin}}(1) = +1$. Exact evaluation of the closed-form drifts.

is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$. The marginal field is affine,

$$b(t, x) = A(t)x + c(t), \quad A = C_t Q^{-1}, \quad c = \dot{m} - A m, \quad (1)$$

with $C_t = \dot{\alpha}\alpha\Sigma_0 + \dot{\sigma}\sigma\Sigma_1$. If $\Sigma_0 = I$, then Q , Q' , A , and Σ_1 share the eigenbasis of Σ_1 , and mode i obeys $x'_i = a_i(t)x_i + c_i(t)$ with $q_i = \alpha^2 + \lambda_i\sigma^2$ and $a_i = q'_i/(2q_i)$. The covariance drift formula is independent of the mean path $c(t)$. We compare the linear path $(1-t, t)$ and trigonometric VP $(\cos(\pi t/2), \sin(\pi t/2))$. Following Chen et al. [2025, Definition 3.2], $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2 dt$. For state-independent A this coincides with their path expectation. By definition $\mathcal{R}$ ignores $c(t)$. Error is Gaussian W_2 of the endpoint laws [Gelbrich, 1990]. This is neither pathwise trajectory error nor local truncation error. Equal NFE uses $S = B/s$ steps with $s = 1, 2, 4$ for Euler, Heun, and RK4.

Any admissible variance path from 1 to $\lambda > 0$ obeys $\int_0^1 a dt = \frac{1}{2} \ln \lambda$. Hence $\mathcal{R} = \int_0^1 a^2 dt = (\int a)^2 + \text{Var}_t(a) \geq (\frac{1}{2} \ln \lambda)^2$, with equality if and only if a is constant. The minimizer is $q(t) = \lambda^t$, $a \equiv \frac{1}{2} \ln \lambda$, which is Chen et al. Example 3.3. In several dimensions, the per-mode schedule $q_i(t) = \lambda_i^t$ yields $\mathcal{R} = (\frac{1}{2} \max_i |\ln \lambda_i|)^2$. It is not in general a shared (α, σ) interpolant.

3 A one-dimensional Gaussian ranking counterexample

Assumption 1. Endpoints $N(0, 1)$ and $N(0, 4)$ are independent. Schedules are linear and trigonometric VP. The solver is explicit Heun with NFE 8 (four steps of size $1/4$).

Proposition 1. Under the assumption, $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1 > \mathcal{R}_{\text{VP}} = \pi^2/16$, while the Heun factors satisfy $r_{\text{lin}} = 6797469/3559400$ and $0 < r_{\text{VP}} < 187/100$. Hence $W_2^{\text{lin}} < 0.091 < 0.130 < W_2^{\text{VP}}$.

Proof sketch. The linear variance is $q_{\text{lin}}(t) = 1 - 2t + 5t^2$. Completing the square gives the equivalent identities $q_{\text{lin}}(t) = 5[(t - 1/5)^2 + (2/5)^2] = 5(t - 1/5)^2 + 4/5$. The substitution $u = t - 1/5 = (2/5) \tan \theta$ converts $\int a_{\text{lin}}^2$ into $(5/2) \int \sin^2 \theta d\theta$. Limits $\arctan(-1/2)$ and $\arctan 2$ together with $\arctan 2 + \arctan(1/2) = \pi/2$ yield $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1$. The VP integral evaluates to $\pi^2/16$ by a Weierstrass identity. The four linear Heun factors are the rationals $47/52, 6/5, 497/370$, and $97/74$, all positive. Exact VP grid values are $a(0) = a(1) = 0$, $a(1/2) = 3\pi/10$, and $a(1/4) = (9 + 15\sqrt{2})\pi/82$, $a(3/4) = (-9 + 15\sqrt{2})\pi/82$. Positivity of $a(3/4)$ is $15^2 \cdot 2 - 9^2 = 369 > 0$. The four-step product lies in $\mathbb{Q}[\pi, \sqrt{2}]$ with nonnegative coefficients, so substituting $\sqrt{2} < 99/70$ and $\pi < 355/113$ is valid and yields $r_{\text{VP}} < 187/100$ by an integer comparison in the anonymous supplement. Interval arithmetic is a cross-check only. $\square$

Table 1: Four schedules: linear, trigonometric VP, shared Chen Ex. 3.3 scalar ($\mathcal{R}_{\text{sc}}$, $M = \lambda_{\text{max}}$), and per-mode log-covariance ($\mathcal{R}_{\text{log}}$). $\mathcal{R}$ is continuous; the last column is the W_2 -minimizer at Heun, NFE 8. Per-mode wins Heun-8 on every geometry; it is not a shared (α, σ) interpolant for $d \geq 2$.

family	d	$\mathcal{R}_{\text{lin}}$	$\mathcal{R}_{\text{VP}}$	$\mathcal{R}_{\text{sc}}$	$\mathcal{R}_{\text{log}}$	Heun-8 W_2 min
anisotropic	2	0.972	0.192	0.173	0.120	log-cov
anisotropic	8	0.972	0.192	0.173	0.120	log-cov
low-rank	2	1.246	0.625	0.668	0.380	log-cov
low-rank	8	2.944	4.731	11.419	2.244	log-cov

Table 2: Strongest centered inversion: low-rank, $d = 8$, Euler, NFE 8. Eigenvalues: 0.05 (six times), 4.64827725629505, 10.675638226546441.

path	$\mathcal{R}$	$\widehat{\mathcal{R}}_{24}$	Gaussian W_2
linear	2.9441044083	2.9476523251	0.8108540111
VP	4.7305438136	4.7295206355	0.4564779075

4 Sign change and transported defects

For any independent linear coupling, $a_{\text{lin}}(0) = -1 < 0 < a_{\text{lin}}(1) = +1$. The VP drift vanishes at the endpoints and does not change sign on $(0, 1)$. Averaging a^2 therefore erases a qualitative distinction that every fixed-stage Runge–Kutta method samples.

Let $e_n = \sqrt{q(t_{n+1})/q(t_n)}$ and let r_n be the numerical factor. The telescoping identity

$$\prod_n r_n - \prod_n e_n = \sum_n (r_n - e_n) \left(\prod_{j < n} r_j \right) \left(\prod_{j > n} e_j \right) \quad (2)$$

is an accounting tool, not the content of the inversion. The content is the leading exact-minus-method coefficient: $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-a''/12 + a^3/6)h^3$ for the implemented Heun factor. Those expansions involve different derivatives, signs, and stage locations.

5 The $\mathcal{R}$ -minimizer and the inversion region

Source $N(0, I)$. Four centered targets: anisotropic Gaussians with condition number 4 in dimensions 2 and 8 (eigenvalues a geometric sequence from $1/2$ to 2); low-rank covariances $FF^\top + 0.05I$ of rank 2 (seed 271828). Because $\mathcal{R}$ uses the spectral norm, the two anisotropic dimensions share the same $\mathcal{R}$ pair (0.972296, 0.192270), so the grid contains three distinct linear-versus-VP $\mathcal{R}$ comparisons. Each geometry is run under Euler, Heun, and RK4 at NFE $\{8, 16, 32\}$. Headline $\mathcal{R}$ is the continuous integral of $\max_i a_i(t)^2$.

The per-mode log-covariance path $q_i(t) = \lambda_i^t$ has the smallest $\mathcal{R}$ on every geometry and the smallest Gaussian W_2 in all 36 blocks (Table 1 reports Heun at NFE 8); it is not a shared (α, σ) interpolant for $d \geq 2$. Linear versus VP invert in 5 of 12 geometry $\times$ solver cells; 4 of those 12 invert at every tested NFE. Trigonometric VP versus the shared Chen Ex. 3.3 scalar schedule ($M = \lambda_{\text{max}}$) invert in 9 of 36 blocks and 4 of 12 cells: anisotropic $d = 2$, Euler at NFE 16 and 32; anisotropic $d = 8$, Euler at NFE 8, 16, and 32; low-rank $d = 2$, Euler at NFE 8 and Heun at NFE 8, 16, and 32; low-rank $d = 8$, none. The fourteen inverted linear-versus-VP solver-budget rows are a hierarchical listing of those cells, not a population rate.

A one-dimensional sweep $\lambda \in [0.05, 100]$ (Figure 2) locates the linear-versus-VP inversions. At $\lambda = 4$, the inversion is present under Heun at every tested budget NFE $\in \{4, 8, 12, 16, 24, 32, 64, 128, 256\}$, absent under Euler, and mixed under RK4 (absent at NFE 8 and 12). Euler *does* invert at more extreme λ on the same sweep. The scalar example is therefore solver-dependent, not an artefact of Heun at budget 8 alone.

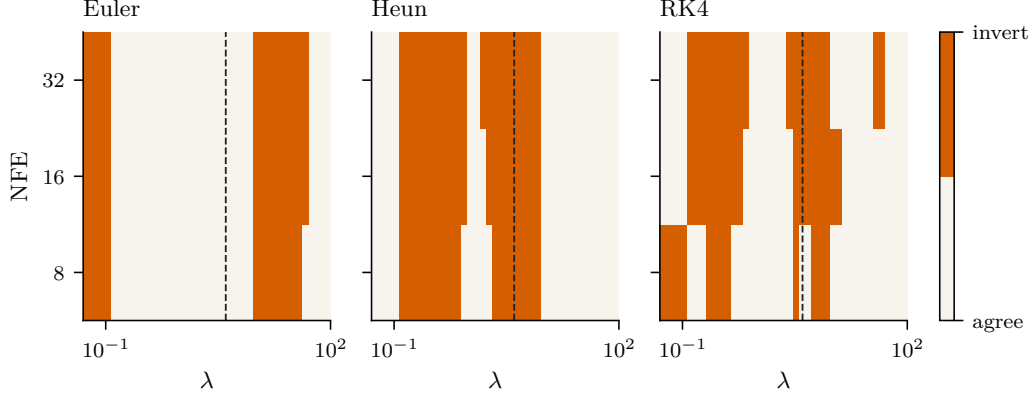


Figure 2: Linear versus VP ranking inversion versus target variance λ . Dark cells: $\mathcal{R}$ and W_2 disagree. The dashed line is $\lambda = 4$ (Proposition 1).

6 Related work and limitations

Flow matching and interpolants supply the Gaussian family [Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023, Tong et al., 2024]. Lipschitz-guided design proposes the selection criterion tested [Chen et al., 2025]. Solver-specific schedules and exact linear parts are documented in diffusion practice [Sabour et al., 2024, Karras et al., 2022, Lu et al., 2022, Zhang and Chen, 2023]. Stéphanovitch [2026] prove sharp Lipschitz regularity of canonical flow-matching fields and Euler-type W_2 rates; that is a bound, not a pairwise ranking of two interpolants at equal NFE.

Tao and Choi [2026] decompose $\nabla v = S + \Omega$ and control Euler error by the logarithmic norm $\mu_2 = \lambda_{\max}(S)$, which is signed, whereas $\mathcal{R}$ integrates $\|A\|_2^2$. In one dimension $\mu_2(a) = a$ and $\|A\|_2 = |a|$. Their Theorem 5 states that McCann displacement interpolation has vanishing material derivative, hence Euler is order 2 in the Eulerian analysis; Remark 7 records the stronger Lagrangian exactness observed on that interpolant. Neither statement applies to the independent-coupling linear or VP path studied here. The present paper supplies a rationally certified Gaussian instance of ranking failure for the unsigned average $\|A\|_2^2$. A counterexample to μ_2 itself is outside our scope.

The systems are commuting affine Gaussian interpolants, three fixed-step Runge–Kutta methods, and small equal-NFE budgets. No learned neural field is evaluated. Chen et al. prove no discretization-error bound and do not claim a universal pairwise ranking at fixed NFE. As an in-family objective, VP versus the shared Ex. 3.3 scalar schedule still invert in 9 of 36 blocks; $\mathcal{R}$ is usable as an objective only when its minimizer is admissible. The Gaussian family is a probe in which $\mathcal{R}$ and W_2 are closed form; in a learned field the linear part is not exact, so exponential integrators that solve an affine scalar mode exactly are not a drop-in substitute. Cell counts are hierarchical descriptive summaries, not population rates.

References

- Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *Journal of Machine Learning Research*, 26(209):1–80, 2025.
- Yifan Chen, Eric Vanden-Eijnden, and Jiawei Xu. Lipschitz-guided design of interpolation schedules in generative models. *arXiv preprint arXiv:2509.01629*, 2025. version 3, 16 May 2026.
- Shukai Du, Junzhe Zhang, and Yiming Li. Lagrangian flow matching: A least-action framework for principled path design. *arXiv preprint arXiv:2605.15419*, 2026.
- Matthias Gelbrich. On a formula for the L^2 Wasserstein metric between measures on Euclidean and Hilbert spaces. *Mathematische Nachrichten*, 147:185–203, 1990.
- Aditi Gupta, Soon Hoe Lim, Annan Yu, and N. Benjamin Erichson. Sharpen your flow: Sharpness-aware sampling for flow matching. *arXiv preprint arXiv:2605.11547*, 2026.
- Ernst Hairer, Syvert P. Nørsett, and Gerhard Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*. Springer, 2nd edition, 1993.
- Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.00364.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.02747.
- Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2209.03003.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. DPM-solver: A fast ODE solver for diffusion probabilistic model sampling in around 10 steps. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.00927.
- Amirmojtaba Sabour, Sanja Fidler, and Karsten Kreis. Align your steps: Optimizing sampling schedules in diffusion models. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2404.14507.
- Arthur Stéphanovitch. Lipschitz regularity in flow matching and diffusion models: Sharp sampling rates and functional inequalities. *arXiv preprint arXiv:2604.06065*, 2026.
- Chenxi Tao and Seung-Kyum Choi. On the role of strain and vorticity in numerical integration error for flow matching. *arXiv preprint arXiv:2605.06680*, 2026.
- Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Hugué, Yanlei Zhang, Jarrod Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024. arXiv:2302.00482.
- Liangyu Yuan, Ruoyu Wang, Tong Zhao, Dingwen Fu, Mingkun Lei, Beier Zhu, and Chi Zhang. Few-step diffusion sampling through instance-aware discretizations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 35882–35892, 2026. arXiv:2603.17671.
- Qinsheng Zhang and Yongxin Chen. Fast sampling of diffusion models with exponential integrator. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2204.13902.